hexadecimal system 24
integer 19
non-positional number system 35
number system 18
octal digit 25
octal system 25
place value 19
positional number system 18
radix 18
real 20
Roman number system 35

2.4.3 SUMMARY

- A number system (or numeral system) is a system that uses distinct symbols to represent a number. In a positional number system, the position a symbol occupies in the number determines the value it represents. Each position has a place value associated with it. A non-positional number system uses a limited number of symbols in which each symbol has a value. However, the position a symbol occupies in the number normally bears no relation to its value: the value of each symbol is normally fixed.
- In the decimal system, the base b = 10 and we use ten symbols to represent numbers. The symbols in this system are often referred to as **decimal digits** or just **digits**. In the binary system, the base b = 2 and we use only two symbols to represent numbers. The symbols in this system are often referred to as **binary digits** or **bits**. In a hexadecimal system, the base = 16 and we use sixteen symbols to represent numbers. The symbols in this system are often referred to as **hexadecimal digits**. In an octal system, the base b = 8 and we use eight symbols to represent numbers. The symbols in this system are often referred to as **octal digits**.
- We can convert a number in any system to decimal. We multiply each digit with its place value in the source system and add the result to get the number in the decimal system. We can convert a decimal number to its equivalent in any base using two different procedures, one for the integral part and one for the fractional part. The integral part needs repeated division and the fraction part needs repeated multiplication.
- Conversion from the binary system to the hexadecimal system and from the hexadecimal system to the binary system is very easy, because four bits in the binary system are represented as one digit in the hexadecimal system.
- Conversion from the binary system to the octal system and from the octal system to the binary system is very easy because three bits in the binary system are represented as one digit in the octal system.

2.5 PRACTICE SET

2.5.1 QUIZZES

A set of interactive quizzes for this chapter can be found on the book's website. It is strongly recommended that the student takes the quizzes to check his/her understanding of the materials before continuing with the practice set.

2.5.2 FILLING-THE-BLANK QUESTIONS

F2-1. The base for a decimal number system is_____

F2-2. The base for an octal number system is_____

F2-3. The base for a binary number system is_____

F2-4. The base for a hexadecimal number system is

F2-4. The base for a hexadecimal number system is_____

F2-5. The binary system uses base_____

F2-6. The decimal system uses base_____

F2-7. The octal system uses base_____

F2-8. The hexadecimal system uses base_____

2.5.3 REVIEW QUESTIONS

Q2-1. Define a number system.

Q2-2. Distinguish between positional and non-positional number systems.

Q2-3. Define the base or radix in a positional number system. What is the relationship between a base and the number of symbols in a positional number system?

Q2-4. Explain the decimal system. Why is it called *decimal?* What is the base in this system?

Q2-5. Explain the binary system. Why is it called *binary?* What is the base in this system?

Q2-6. Explain the octal system. Why is it called *octal?* What is the base in this system?

Q2-7. Explain the hexadecimal system. Why is it called *hexadecimal?* What is the base in this system?

Q2-8. Why is it easy to convert from binary to hexadecimal and *vice versa?*

Q2-9. How many bits in the binary system are represented by one digit in the hexadecimal system?

Q2-10. How many bits in the binary system are represented by one digit in the octal system?

2.5.4 PROBLEMS

P2-1. Convert the following binary numbers to decimal without using a calculator, showing your work:

- **a.** $[01101)_2$
- **b.** $(1011000)_2$
- **c.** $(011110.01)_2$
- **d.** $(111111.111)_2$

P2-2. Convert the following hexadecimal numbers to decimal without using a calculator, showing your work:

- **a.** $(AB2)_{16}$
- **b.** $(123)_{16}$
- **c.** $(ABB)_{16}$
- **d.** $(35E.E1)_{16}$

P2-3. Convert the following octal numbers to decimal without using a calculator, showing your work:

- **a.** $(237)_g$
- **b.** $(273\ l)_g$
- **c.** $(617.7)_g$
- **d.** $(21.11)_g$

P2-4. Convert the following decimal numbers to binary without using a calculator, showing your work:

calculator, showing your work:

a. 1234
b. 88
c. 124.02
d. 14.56

P2-5. Convert the following decimal numbers to octal without using a calculator, showing your work:

a. 1156
b. 99
c. 11.4
d. 72.8

P2-6. Convert the following decimal numbers to hexadecimal without using a calculator, showing your work:

a. 567
b. 1411
c. 12.13
d. 16

P2-7. Convert the following octal numbers to hexadecimal without using a calculator, showing your work:

a. $(514)_g$
b. $(411)_g$
c. $(13.7)_g$
d. $(1256)_g$

P2-8. Convert the following hexadecimal numbers to octal without using a calculator, showing your work:

a. $(51A)_{16}$
b. $(4E1)_{16}$
c. $(BB.C)_{16}$
d. $(ABCD)_{16}$

P2-9. Convert the following binary numbers to octal without using a calculator, showing your work:

a. $(01101)_2$
b. $(1011000)_2$
c. $(011110.01)_2$
d. $(111111.111)_2$

P2-10. Convert the following binary numbers to hexadecimal without using a calculator, showing your work:

a. $(01101)_2$
b. $(1011000)_2$
c. $(011110.01)_2$

d. $(111111.111)_2$

P2-11. Convert the following decimal numbers to binary using the alternative method discussed in Example 2.28, showing your work:

a. 121
b. 78
c. 255
d. 214

P2-12. Change the following decimal numbers into binary using the alternative method discussed in Example 2.29, showing your work:

a. $3^5/_8$
b. $12^3/_{32}$
c. $4^{13}/_{64}$
d. $12^5/_{128}$

P2-13. In a positional number system with base b, the largest integer number that can be represented using *k* digits is b^k - 1. Find the largest number in each of the following systems with *six* digits:

a. Binary
b. Decimal
c. Hexadecimal
d. Octal

P2-14. Without converting, find the minimum number of digits needed in the destination system for each of the following cases:

a. Five-digit decimal number converted to binary.
b. Four-digit decimal converted to octal.
c. Seven-digit decimal converted to hexadecimal.

P2-15. Without converting, find the minimum number of digits needed in the destination system for each of the following cases:

a. 5-bit binary number converted to decimal.
b. Three-digit octal number converted to decimal.
c. Three-digit hexadecimal converted to decimal.

P2-16. The following table shows how to rewrite a fraction so the denominator is a power of two (1, 4, 8, 16, and so on).

Original	*New*	*Original*	*New*
0.5	½	0.25	¼
0.125	$1/_8$	0.0625	$1/_{16}$
0.03125	$1/_{32}$	0.015625	$1/_{64}$

However, sometimes we need a combination of entries to find the appropriate fraction. For example, 0.625 is not in the table, but we know that 0.625 is 0.5 + 0.125. This means that 0.625 can be written as ½ + ⅛, or ⅝.

Change the following decimal fractions to a fraction with a power of 2.

a. 0.1875

b. 0.640625
c. 0.40625
d. 0.375

P2-17. Using the results of the previous problem, change the following decimal numbers to binary numbers.

a. 7.1875
b. 12.640625
c. 11.40625
d. 0.375

P2-18. Find the maximum value of an integer in each of the following cases:

a. b = 10, k = 10
b. b = 2, k = 12
c. b = 8, k = 8
d. b = 16, k = 7

P2-19. Find the minimum number of required bits to store the following integers:

a. less than 1000
b. less than 100000
c. less than 64
d. less than 256

P2-20. A number less than b^k can be represented using k digits in base b. Show the number of digits needed in each of the following cases.

a. Integers less than 2^{14} in binary
b. Integers less than 10^8 in decimal
c. Integers less than 8^{13} in octal
d. Integers less than 16^4 in hexadecimal

P2-21. A common base used on the internet is b = 256. We need 256 symbols to represent a number in this system. Instead of creating this large number of symbols, the designers of this system have used decimal numbers to represent a symbol: 0 to 255. In other words, the set of symbols is S = {0, 1, 2, 3, ..., 255}. A number in this system is always in the format $S_rS_2.S_3.S_4$ with four symbols and three dots that separate the symbols. The system is used to define internet addresses (see Chapter 6). An example of an address in this system is 10.200.14.72, which is equivalentto $10 \times 256^3 + 200 \times 256^2 + 14 \times 256^1 + 72 \times 256^0 = 180883016$ in decimal. This number system is called *dotted decimal notation.* Find the decimal value of each of the following internet addresses:

a. 17.234.34.14
b. 14.56.234.56
c. 110.14.56.78
d. 24.56.13.11

P2-22. Internet addresses described in the previous problem are also represented as patterns of bits. In this case, 32 bits are used to represent an address, eight bits for each symbol in dotted decimal notation. For example, the

address 10.200.14.72 can also be represented as 00001010 11001000 00001110 01001000. Show the bit representation of the following internet addresses:

- **a.** 17.234.34.14
- **b.** 14.56.234.56
- **c.** 110.14.56.78
- **d.** 24.56.13.11

P2-23. Write the decimal equivalent of the following Roman numbers:

- **a.** XV
- **b.** XXVII
- **c.** VLIII
- **d.** MCLVII

P2-24. Convert the following decimal numbers to Roman numbers:

- **a.** 17
- **b.** 38
- **c.** 82
- **d.** 999

P2-25. Find which of the following Roman numerals are not valid:

- **a.** MMIM
- **b.** MIC
- **c.** CVC
- **d.** VX

P2-26. Mayan civilization invented a positional vigesimal (base 20) numeral system, called the *Mayan numeral system*. They use base 20 probably because they used both their fingers and toes for counting. This system has 20 symbols that are constructed from three simpler symbols. The advanced feature of the system is that it has a symbol for zero, which is a shell. The other two symbols are a circle (or a pebble) for one and a horizontal bar (or a stick) for five. To represent a number greater than nineteen, numerals are written vertically. Search the internet to answer the following: what are the decimal numbers 12, 123, 452, and 1256 in the Mayan numeral system?

P2-27. The *Babylonian* civilization is credited with developing the first positional numeral system, called the *Babylonian numeral system*. They inherited the Sume-rian and Akkadian numeral system and developed it into positional sexagesimal system (base 60). This base is still used today for times and angles. For example, one hour is 60 minutes and one minute is 60 seconds: similarly, one degree is 60 minutes and one minute is 60 seconds. As a positional system with base b requires b symbols (digits), we expect a positional sexagesimal system to require 60 symbols. However, the Babylonians did not have a symbol for zero, and produced the other 59 symbols by stacking two symbols, those for one and ten. Search the internet to answer the following questions:

- **a.** Express the following decimal numbers in Babylonian numerals: 11 291, 3646, 3582.
- **b.** Mention problems that might arise from not having a symbol for 0. Find how the Babylonian numeral system addresses the problem.